

Weight perturbation across four AI weather models

Multiplicative weight noise $\tilde{W}_m = W \odot (1 + \sigma \xi_m)$, $\xi_m \sim \mathcal{N}(0, 1)$, independent across members.

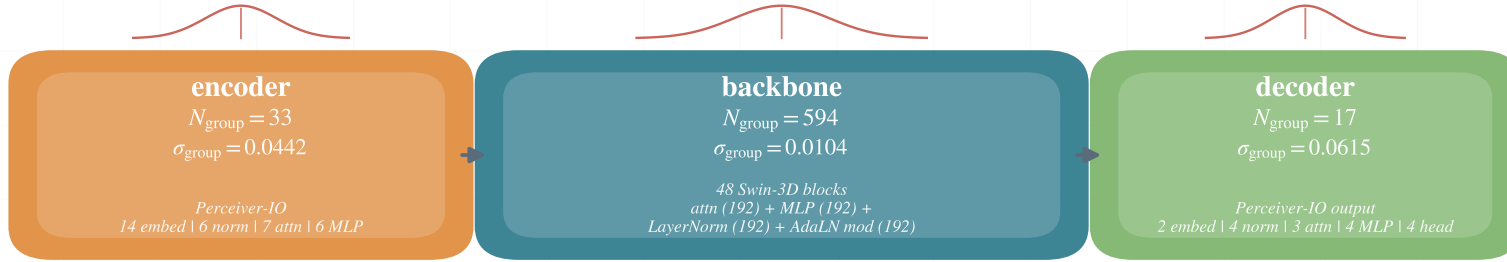
Phase 1: perturb all weights, sweep magnitude $\sigma \in \{0.001, 0.003, 0.01, 0.03, 0.1\}$. Phase 2: one layer group, $\sigma_{\text{group}} = \sigma_{\text{full}} \sqrt{N_{\text{total}}/N_{\text{group}}}$ (shown here at $\sigma_{\text{full}} = 0.01$).



Aurora

Swin-Transformer 3D U-Net

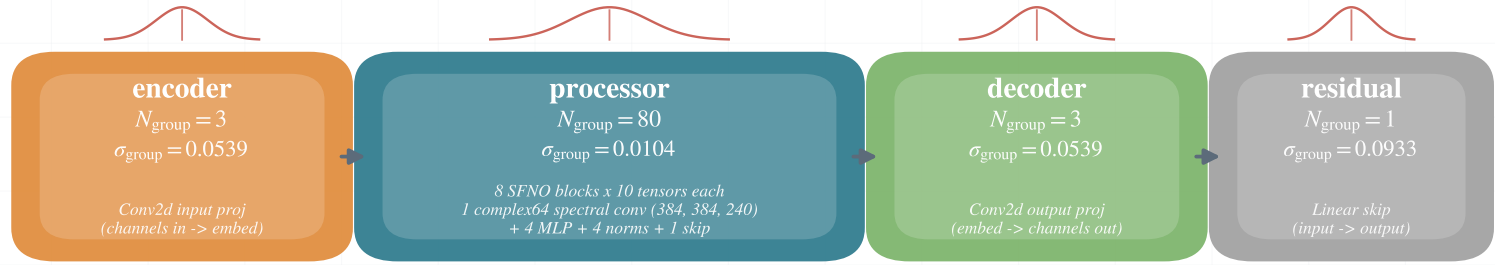
$N_{\text{total}} = 644$



SFNO

Spherical Fourier Neural Operator

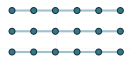
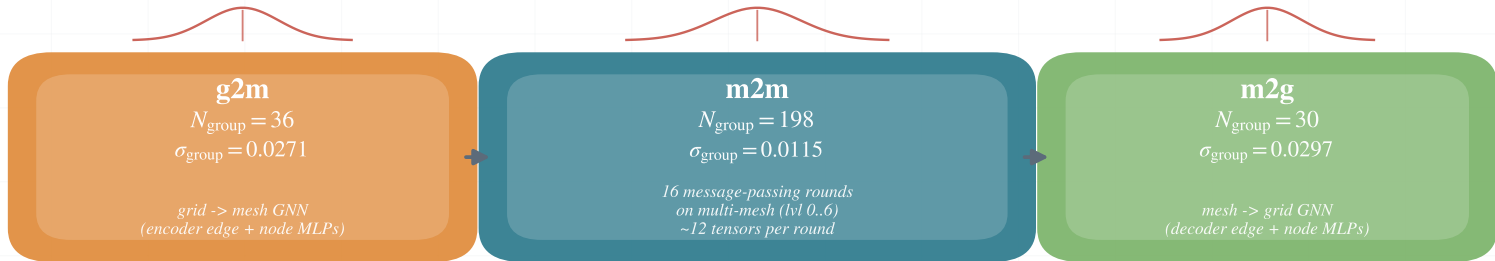
$N_{\text{total}} = 87$



GraphCast

Multi-Mesh Graph Neural Network

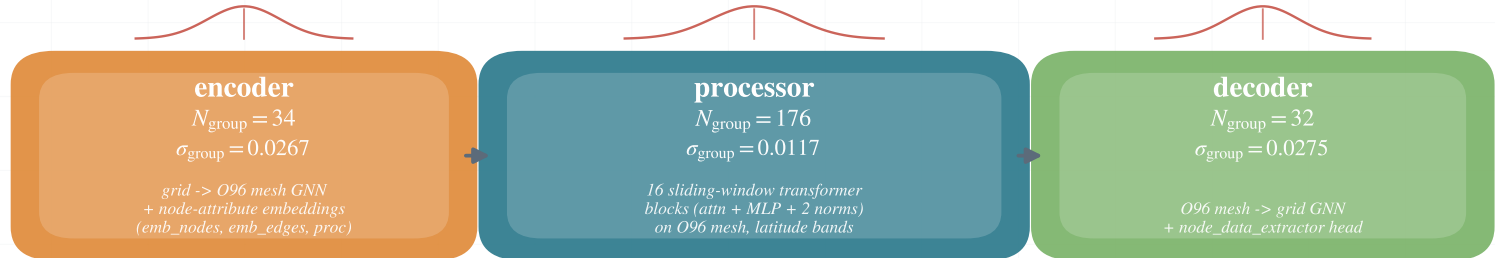
$N_{\text{total}} = 264$



AIFS

GNN + latitude-band transformer

$N_{\text{total}} = 242$



encoder

processor

decoder

aux / residual



= Gaussian noise injection on every learnable tensor in the group